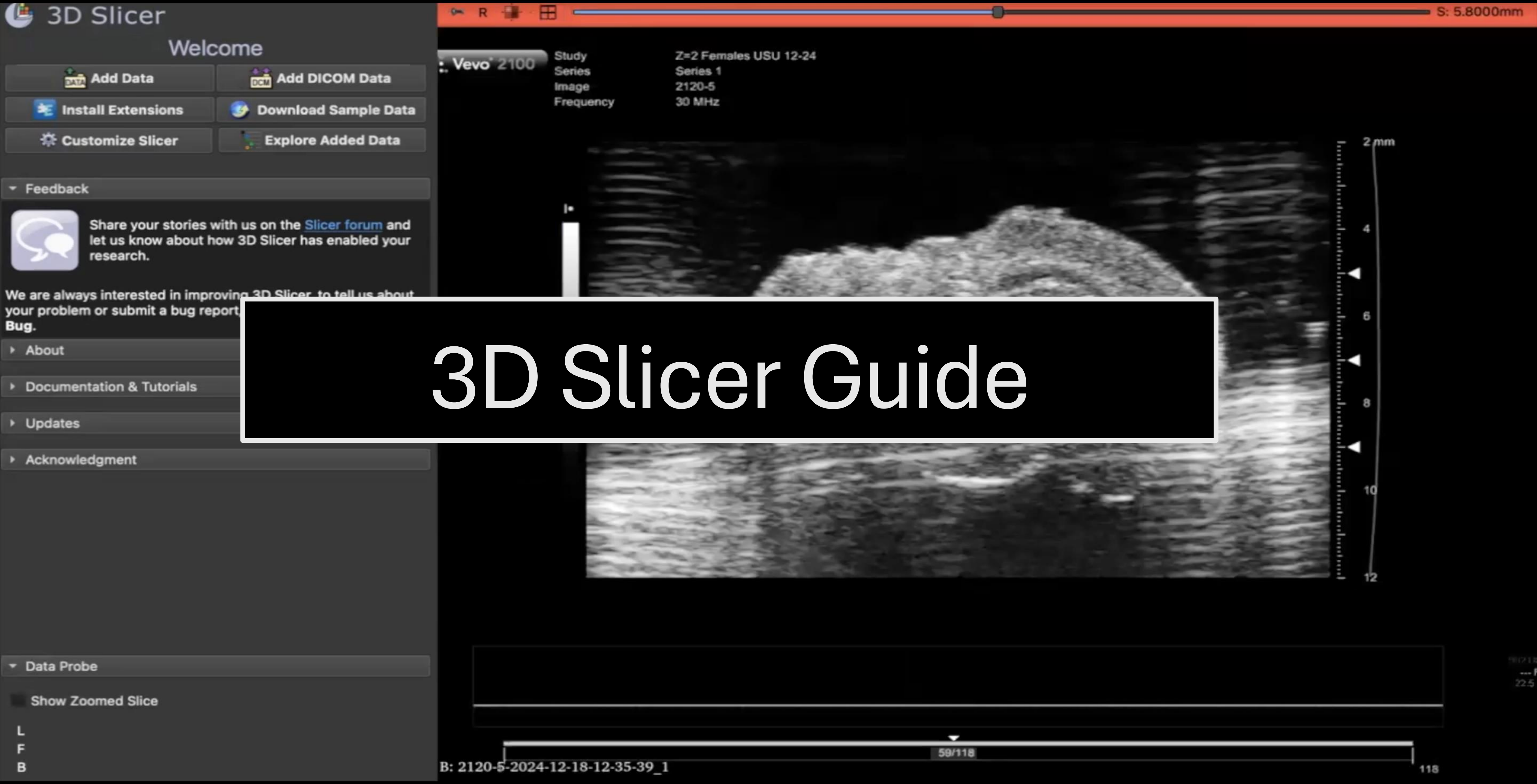
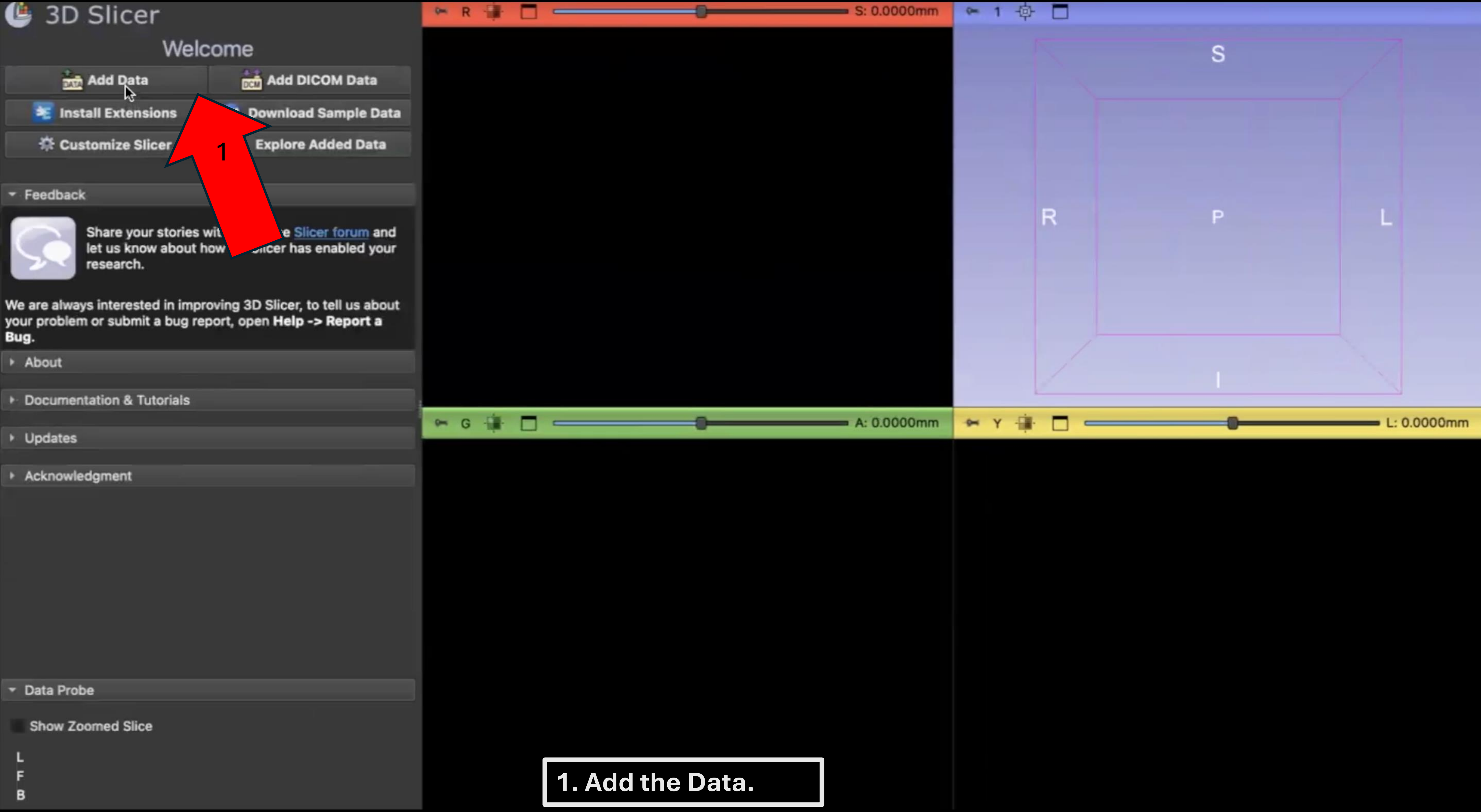
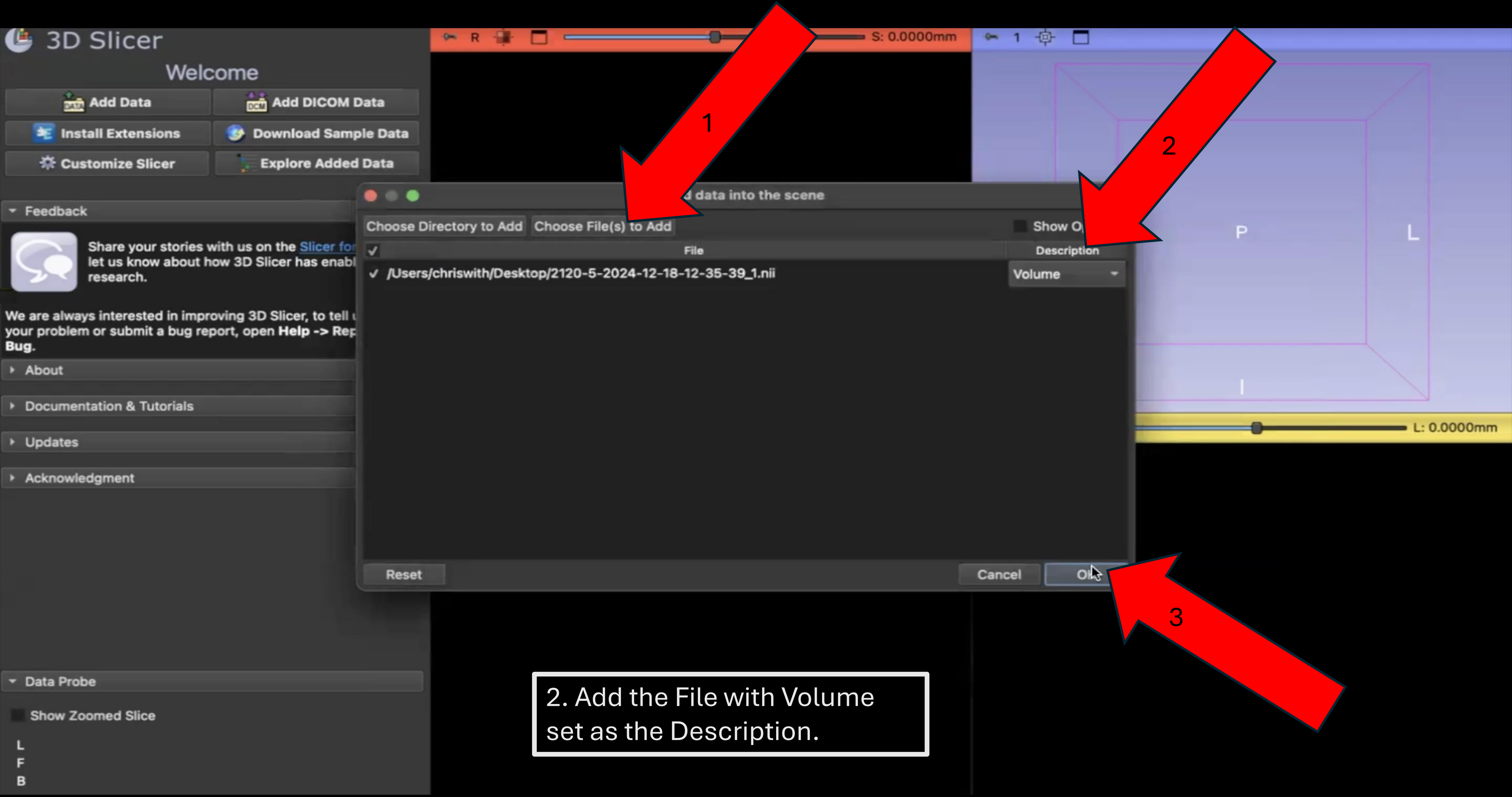
3D Slicer Guide

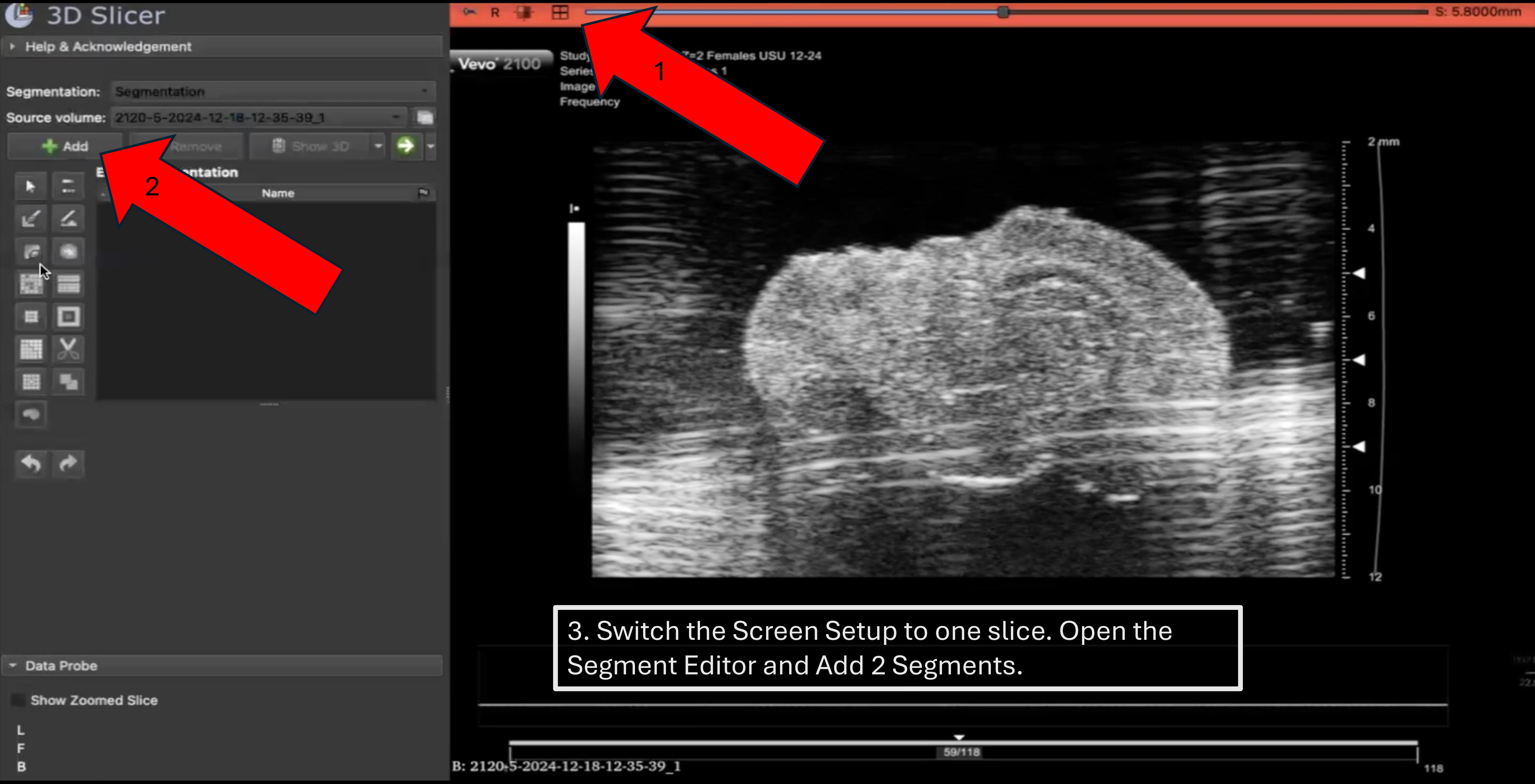




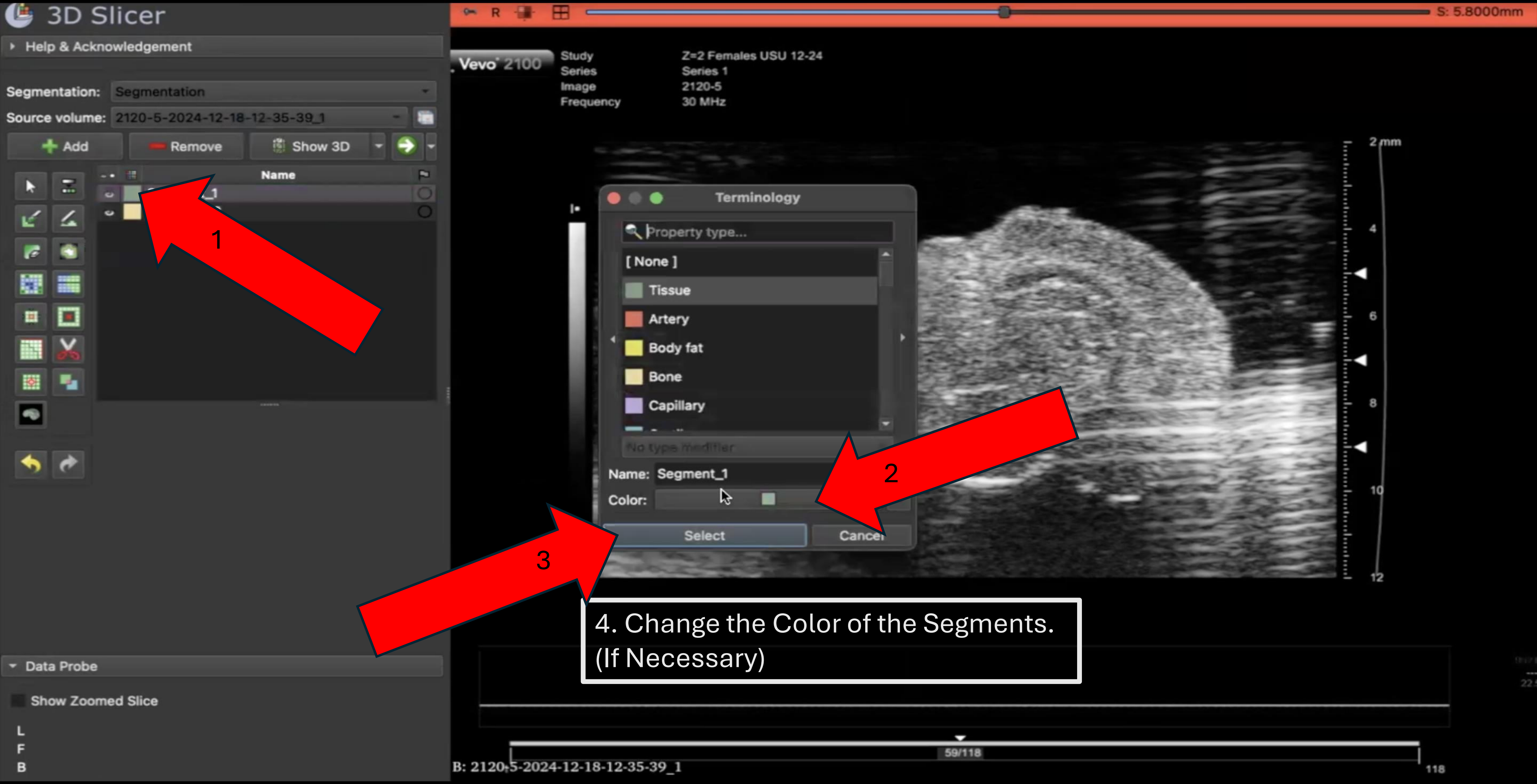
1. Add the Data.



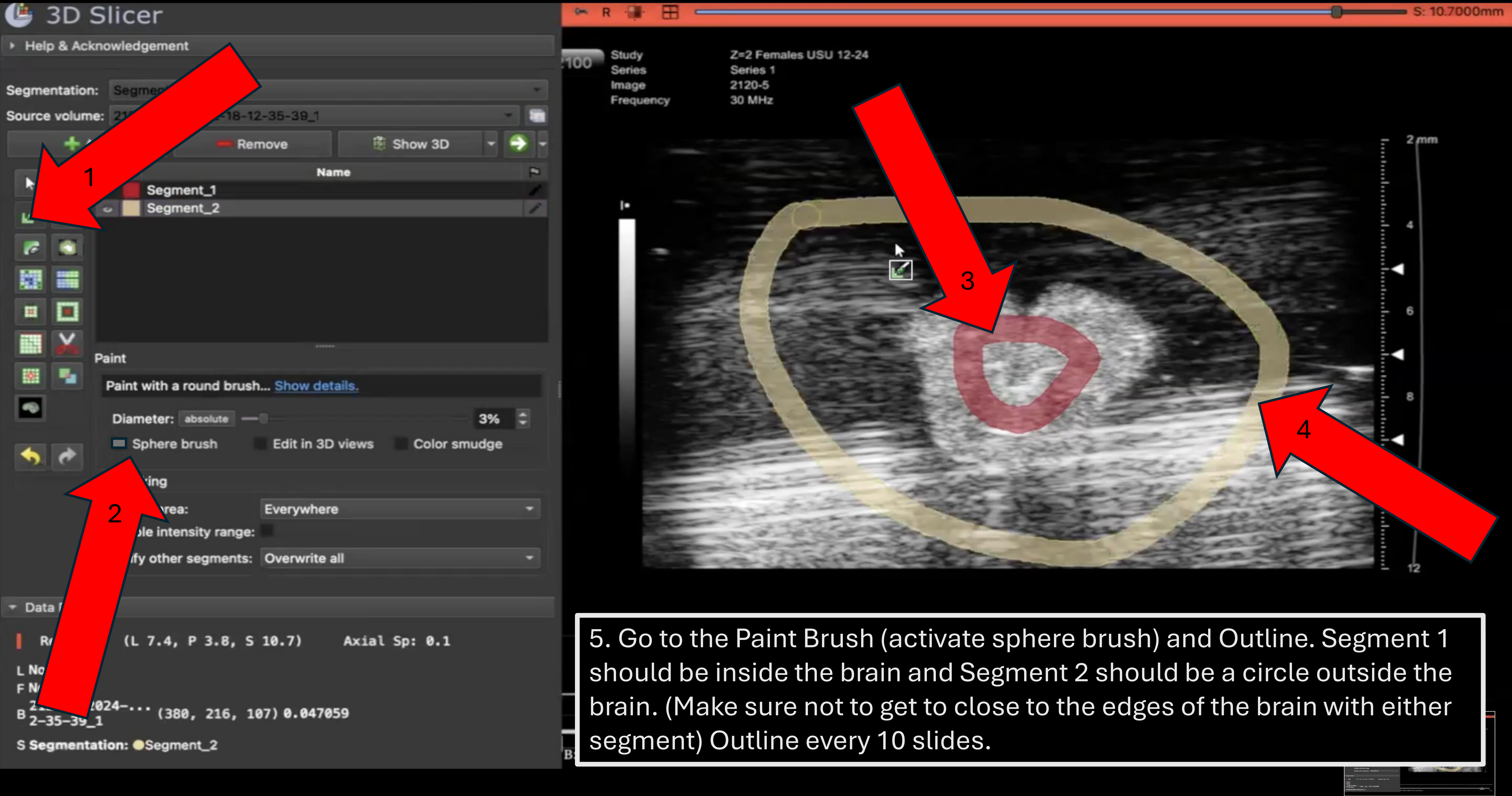
2. Add the File with Volume set as the Description.



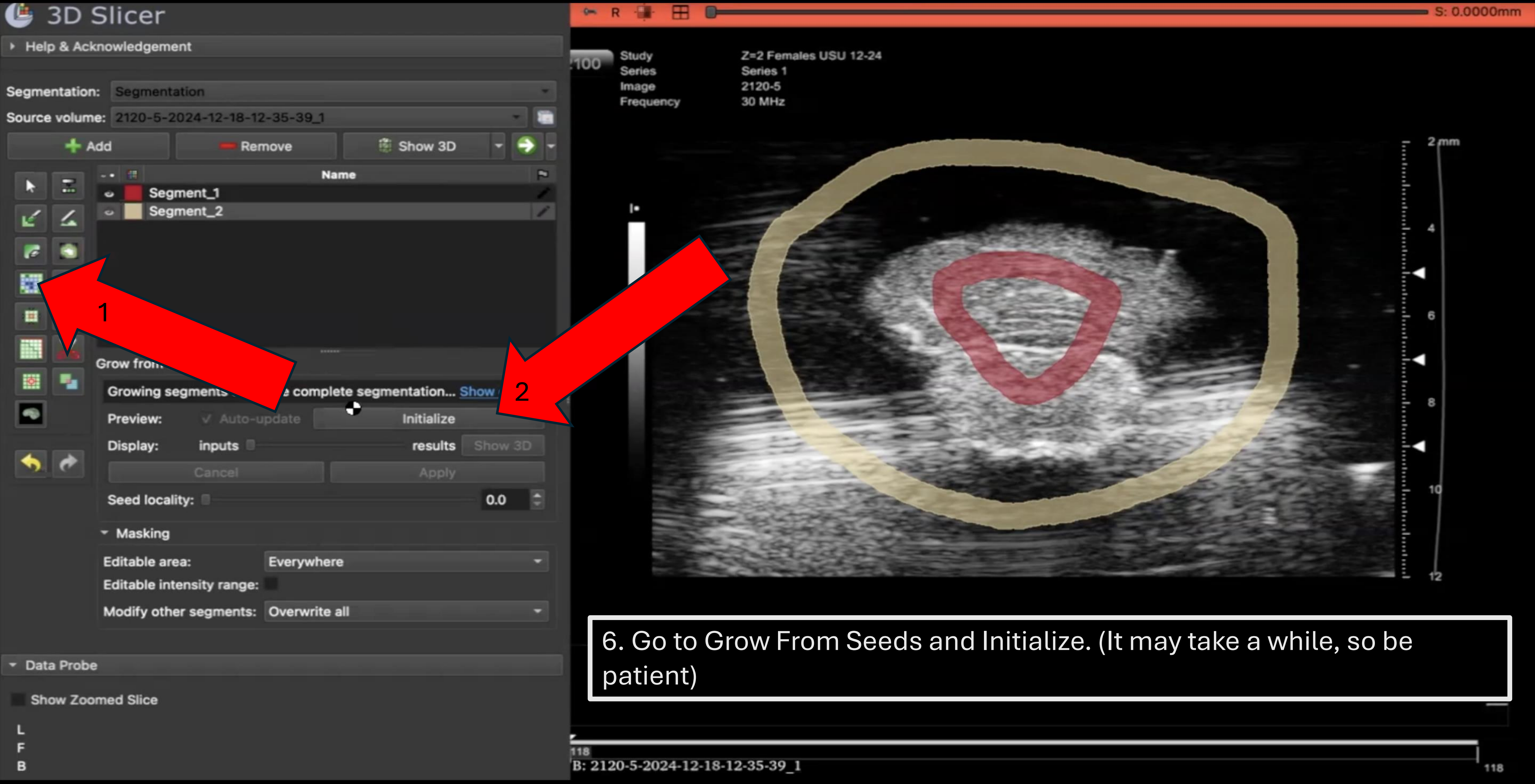
3. Switch the Screen Setup to one slice. Open the Segment Editor and Add 2 Segments.



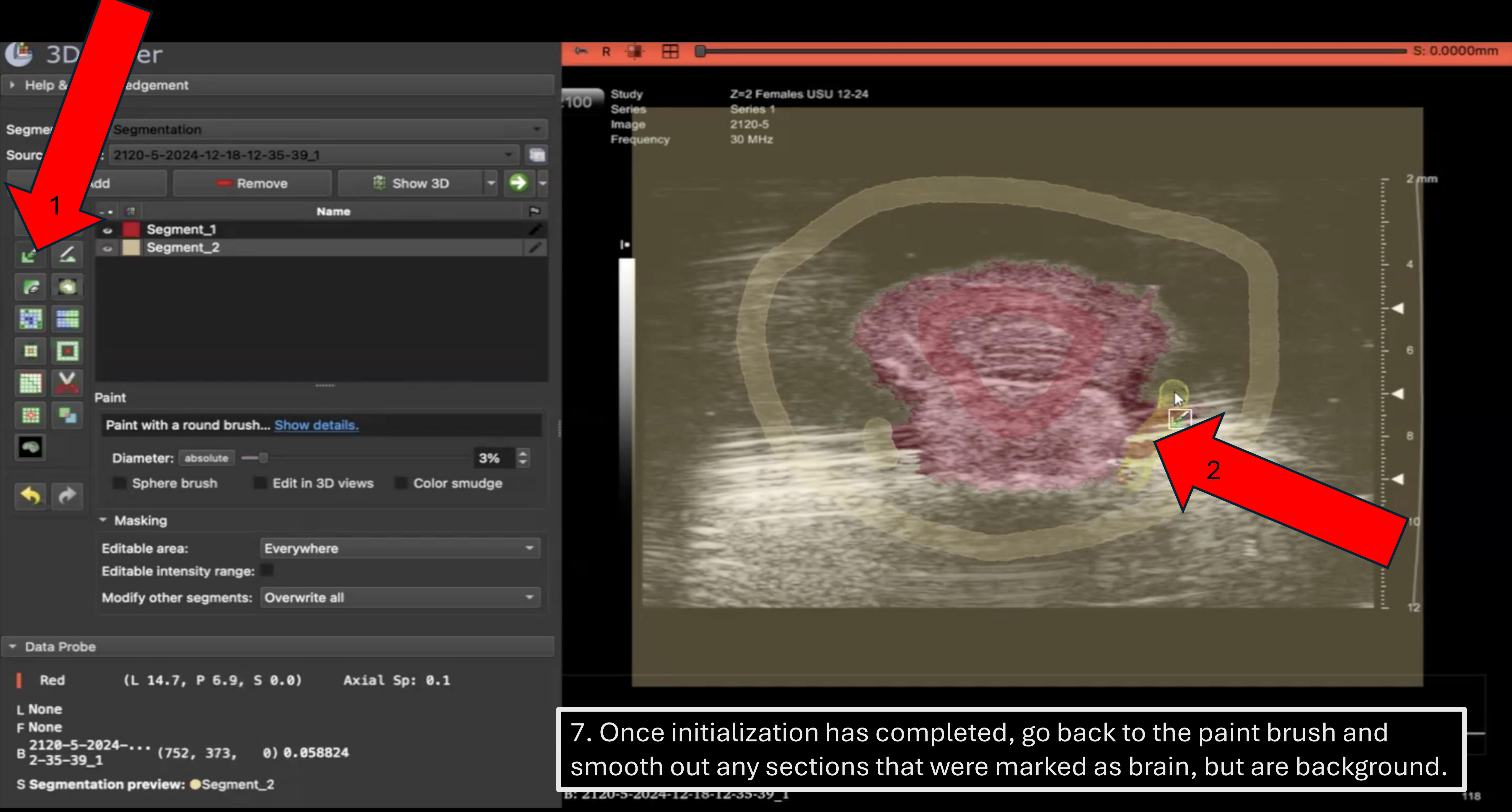
4. Change the Color of the Segments. (If Necessary)



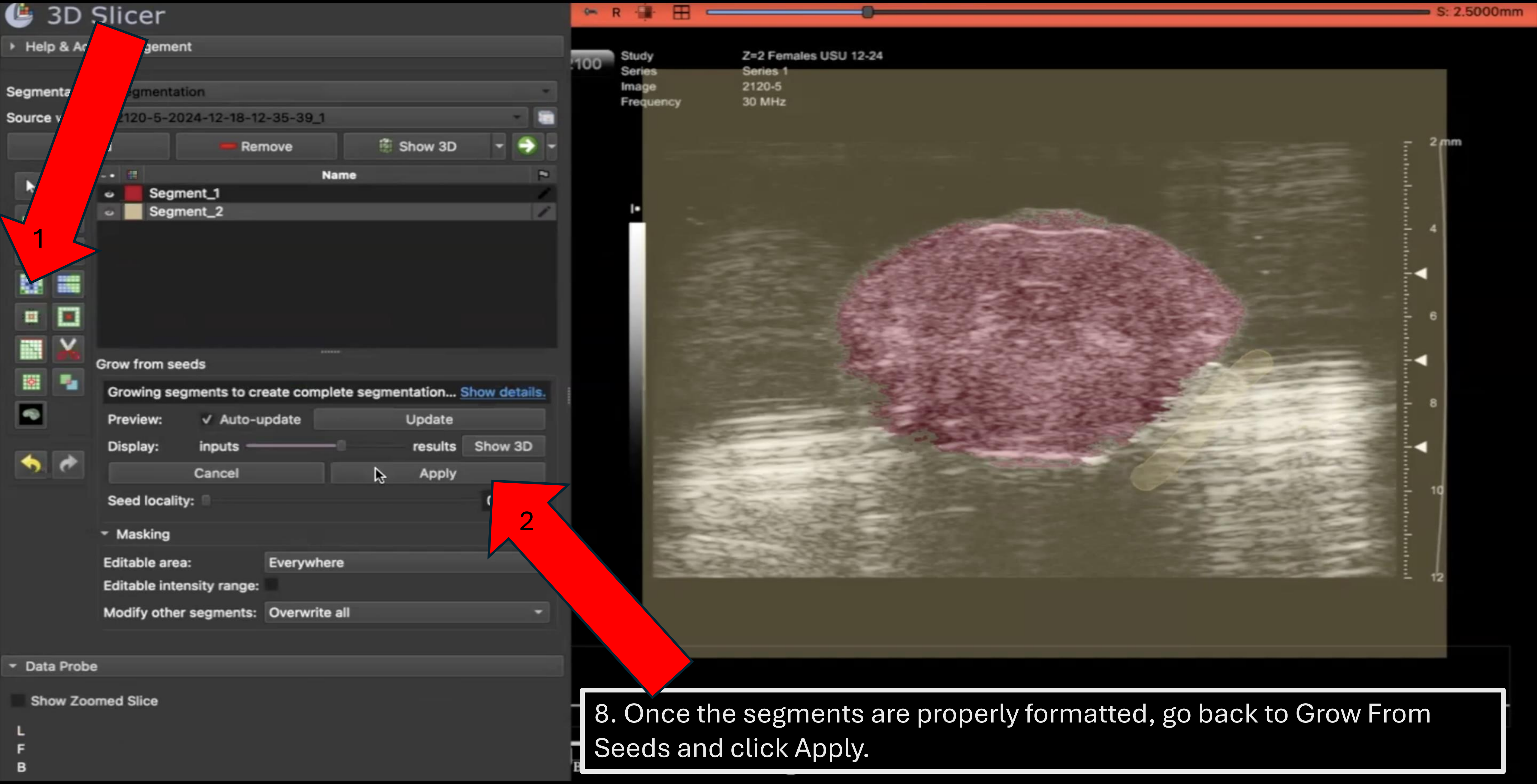
5. Go to the Paint Brush (activate sphere brush) and Outline. Segment 1 should be inside the brain and Segment 2 should be a circle outside the brain. (Make sure not to get to close to the edges of the brain with either segment) Outline every 10 slides.



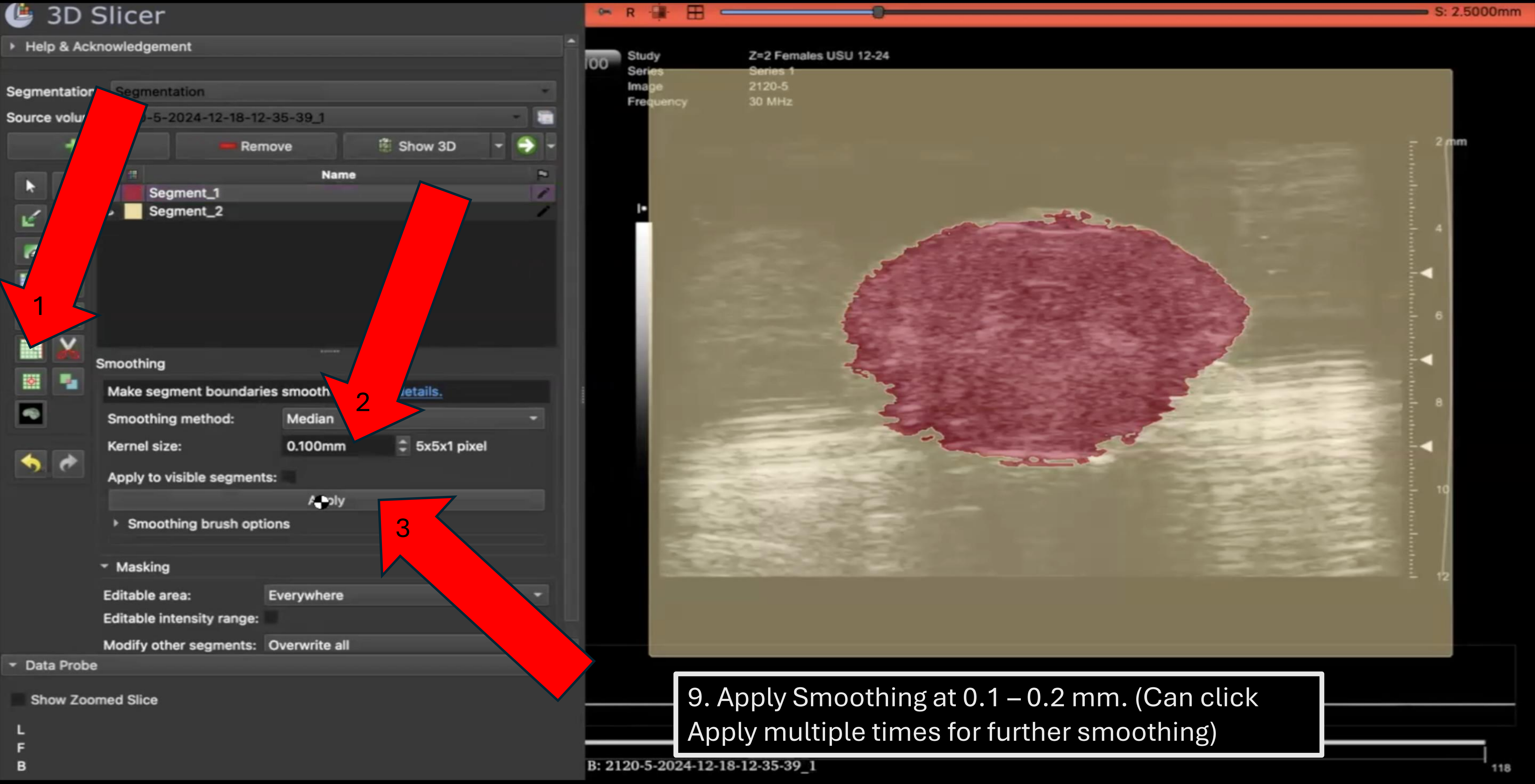
6. Go to Grow From Seeds and Initialize. (It may take a while, so be patient)



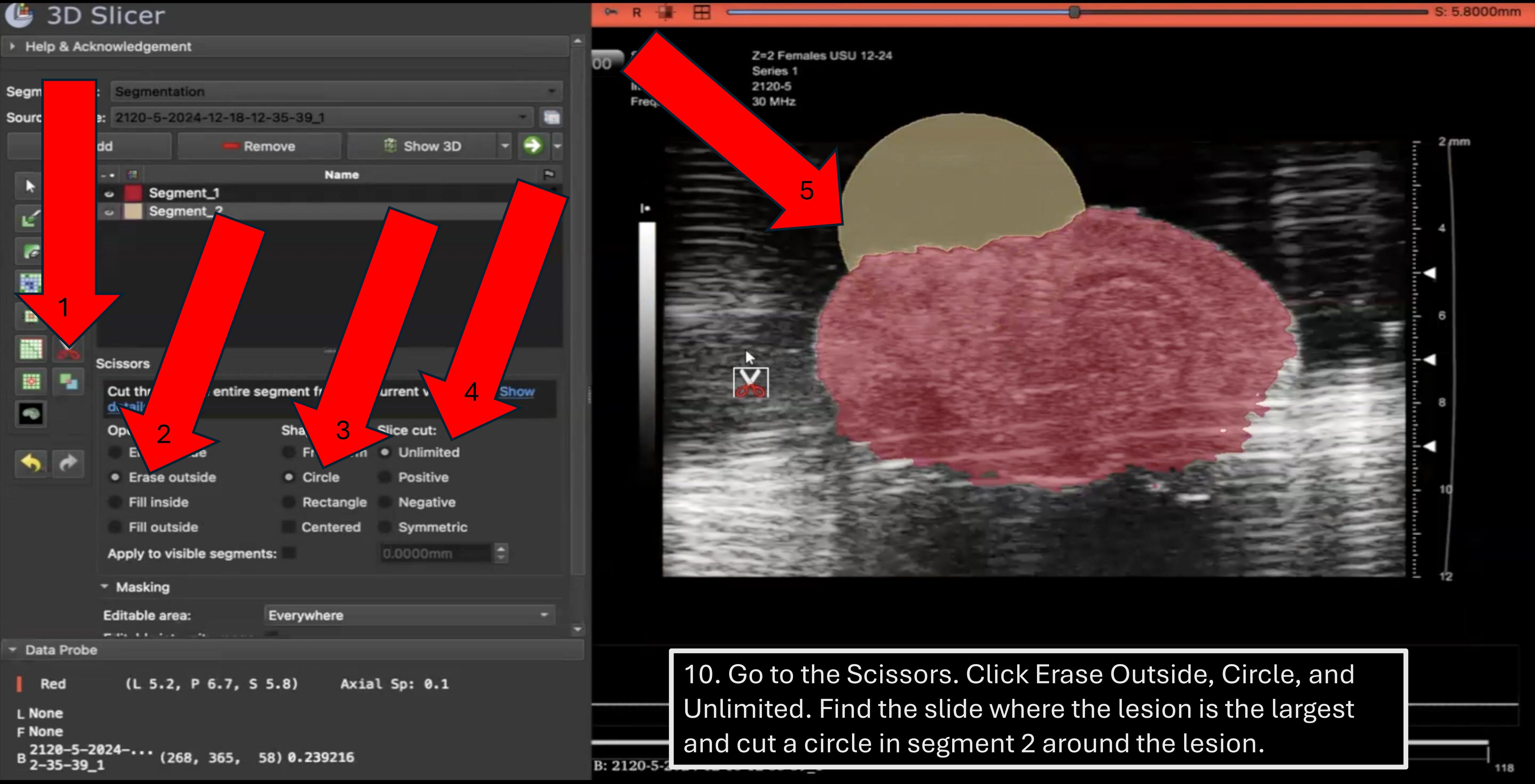
7. Once initialization has completed, go back to the paint brush and smooth out any sections that were marked as brain, but are background.



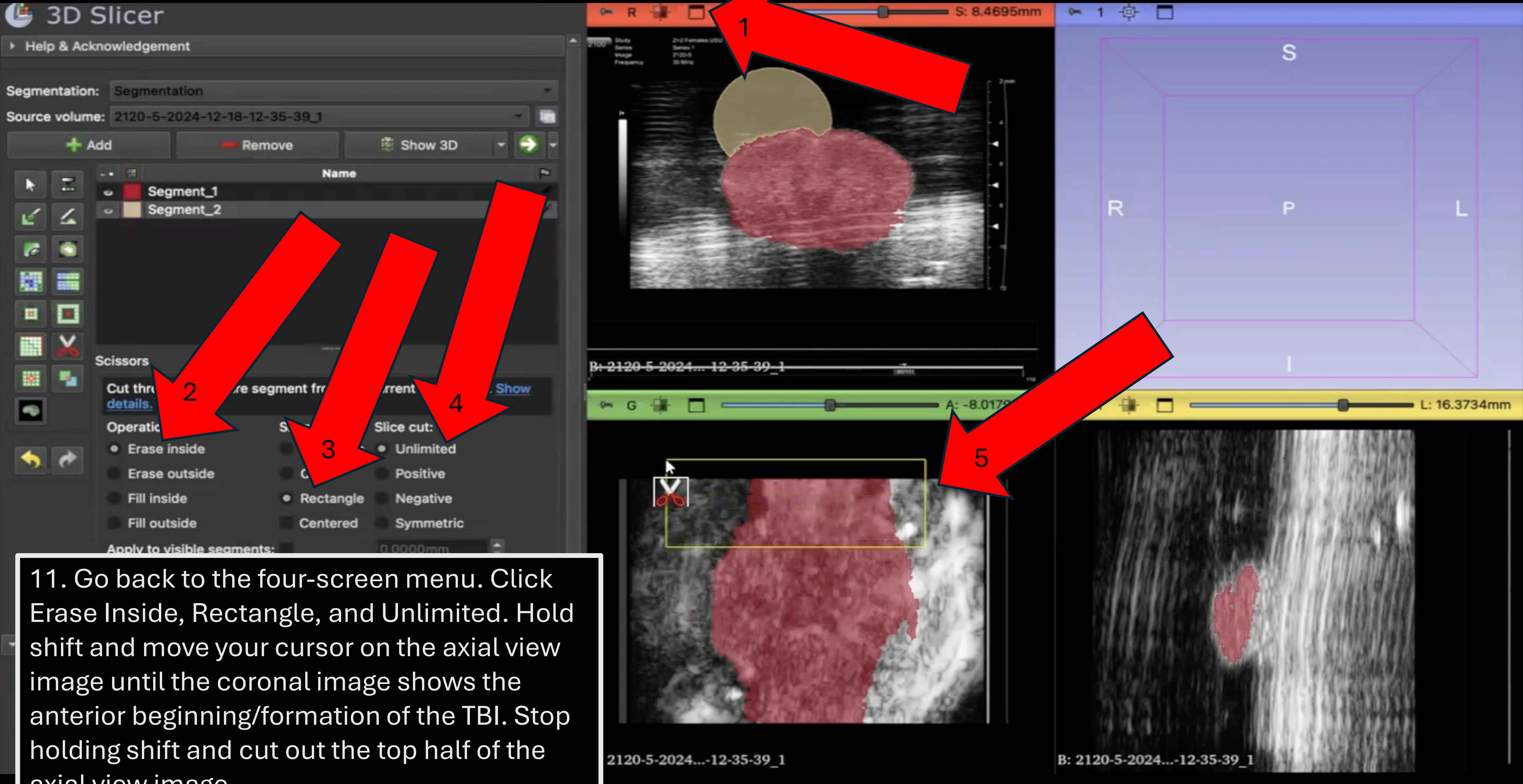
8. Once the segments are properly formatted, go back to Grow From Seeds and click Apply.



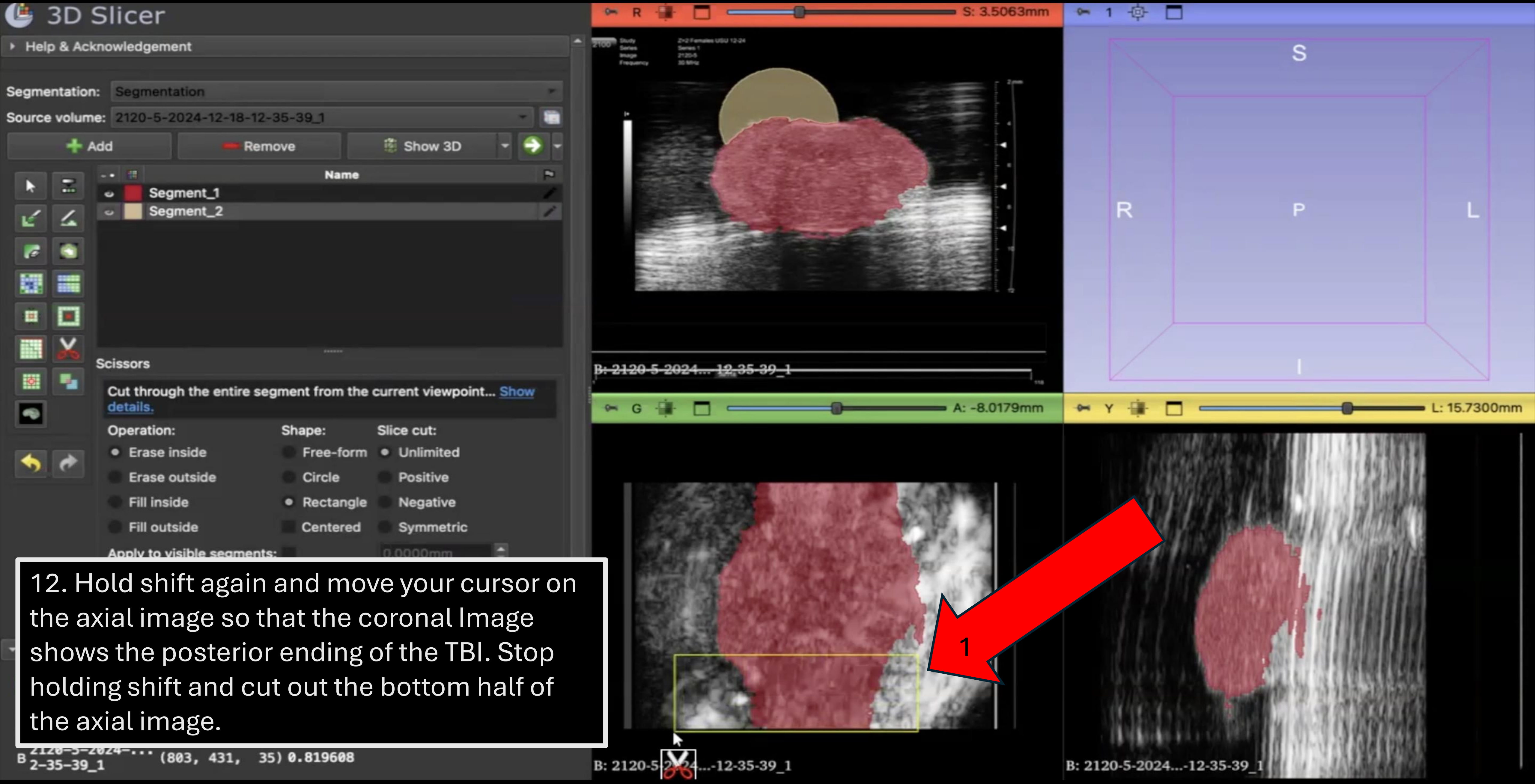
9. Apply Smoothing at 0.1 – 0.2 mm. (Can click Apply multiple times for further smoothing)



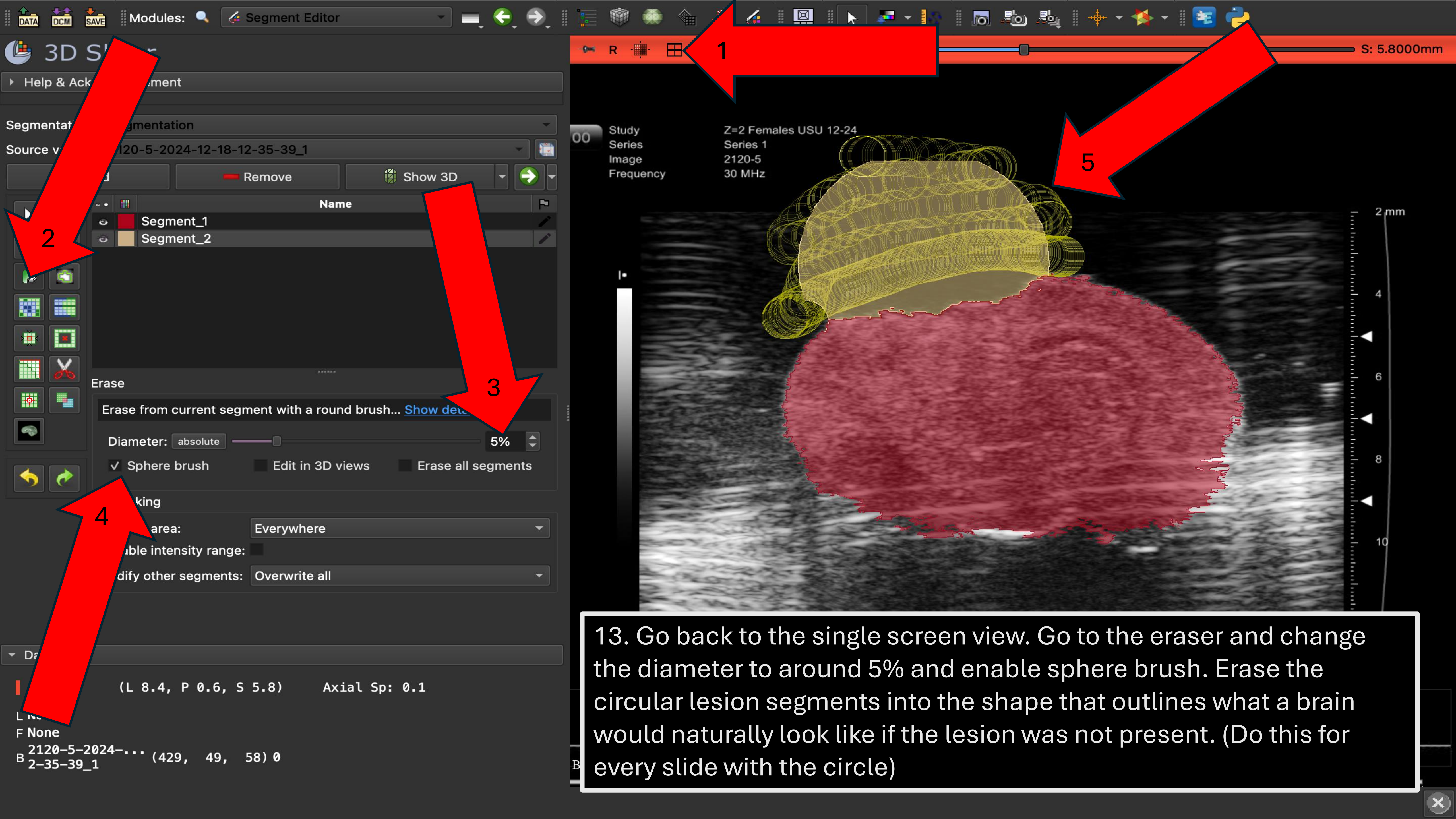
10. Go to the Scissors. Click Erase Outside, Circle, and Unlimited. Find the slide where the lesion is the largest and cut a circle in segment 2 around the lesion.



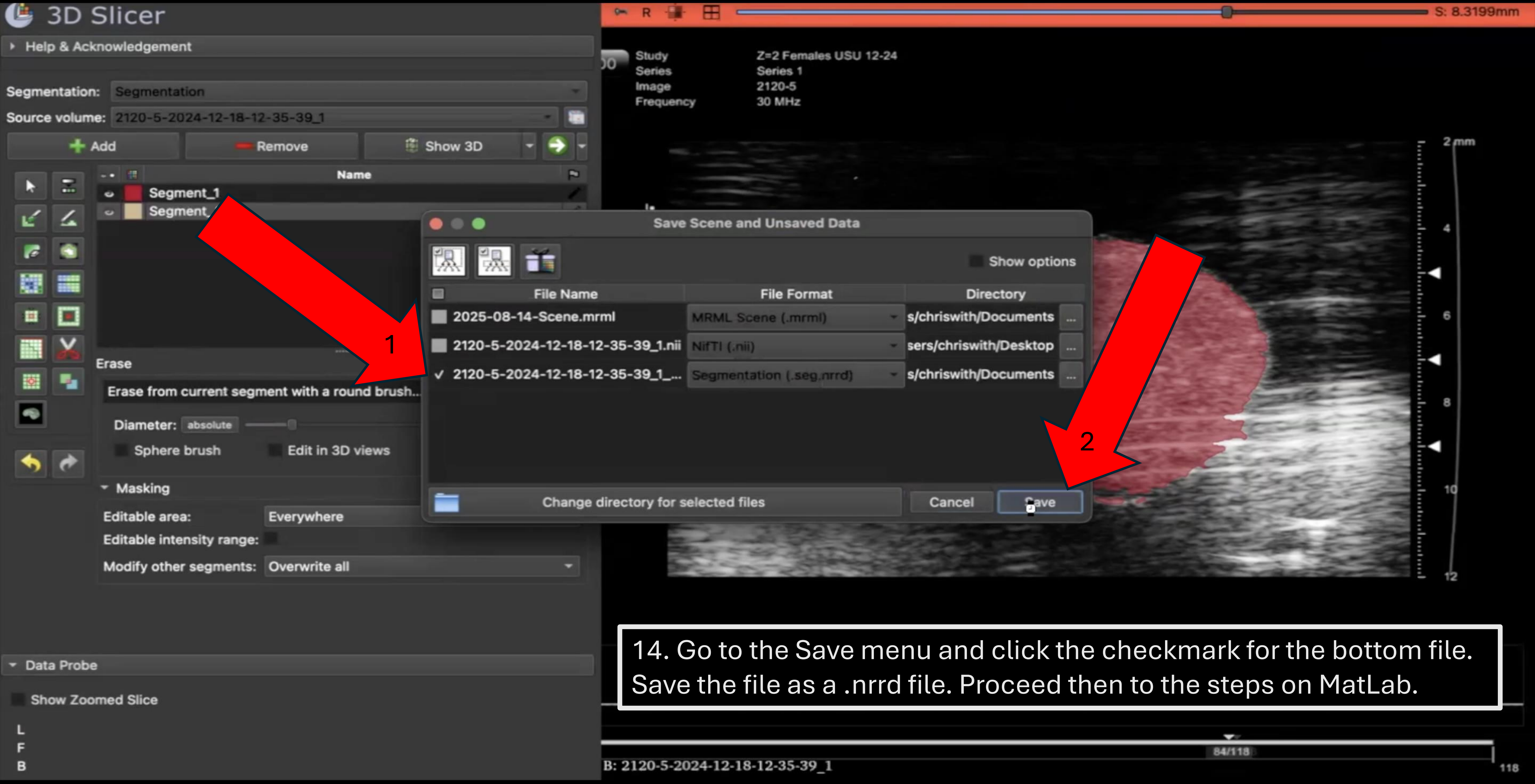
11. Go back to the four-screen menu. Click Erase Inside, Rectangle, and Unlimited. Hold shift and move your cursor on the axial view image until the coronal image shows the anterior beginning/formation of the TBI. Stop holding shift and cut out the top half of the axial view image.



12. Hold shift again and move your cursor on the axial image so that the coronal Image shows the posterior ending of the TBI. Stop holding shift and cut out the bottom half of the axial image.



13. Go back to the single screen view. Go to the eraser and change the diameter to around 5% and enable sphere brush. Erase the circular lesion segments into the shape that outlines what a brain would naturally look like if the lesion was not present. (Do this for every slide with the circle)



14. Go to the Save menu and click the checkmark for the bottom file. Save the file as a .nrrd file. Proceed then to the steps on MatLab.